NAME: ______________________________ **PERIOD:** ________ **DATE:** ____________

Two Measurements per Technician

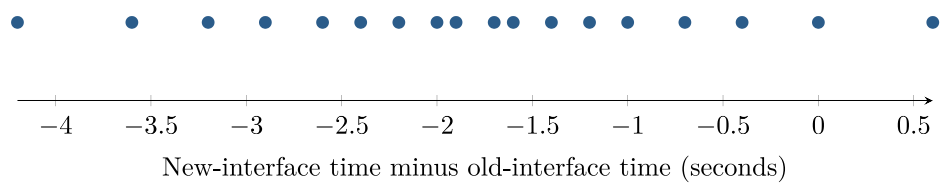
AP Statistics · Topic 7.4 (CED 4.4) · B: 2027-01-19 · E: 2027-03-05

[STAR] TODAY'S PROBLEM

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows $d = \text{new time} - \text{old time}$. Researchers ask whether the new interface reduces the population mean time.

Nia: "The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail."

Colin: "There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative."



FIRST TAKE — before the video (5 min)

Which setup better aligns with the study? Cite the measurement structure or the direction of the claim.

[BOOK] PAIR THE DATA BEFORE YOU TEST (keep on the page)

When the same individual is measured twice, define an order of subtraction and analyze the resulting differences as one quantitative sample. For $d = \text{new} - \text{old}$, a reduction corresponds to $H_a : \mu_d < 0$; the null is $H_0 : \mu_d = 0$. With unknown population standard deviation, use a matched-pairs one-sample t-test on the differences. Verify random sampling or random assignment, $n \leq 0.10N$ when sampling without replacement, and an approximately normal sampling distribution. If $n < 30$, inspect the differences themselves for strong skewness and outliers.

Finish after the video:

DOK 1 **(a)** Define μ_d and state hypotheses for the research question.

DOK 2 **(b)** Calculate $\bar{d}$ and s_d, verify random, 10 percent, and shape conditions, and name the procedure.

DOK 3 **(c)** Adjudicate the setups using the same 18 technicians, subtraction order, directional question, and difference plot. Explain why 36 times are not 36 independent units and how the rejected analysis could change both uncertainty and the question answered.

Frame: The independent units are ____________________; preserving each pair changes the analysis by ____________________________________.

Frame: With the stated subtraction order, the claim corresponds to _________________; the rejected setup instead answers ____________________________________.

EXIT — turn this in

One thing the video changed about my first take: ____